

UNIT 1: Computer Fundamentals

Marks:10

Computer: A computer is an electronic device that accepts data as input, stores data and instructions, processes the data, and finally returns the result to the user. A computer system is a collection of hardware and software components. The term 'Computer' is derived from the word 'Compute', which means to calculate. It performs arithmetic and logical operations. It can process large volumes of data with high speed, accuracy and reliability.

Applications of computers:

Computer system is used in every walk of life. The wide availability of the internet has enhanced the use of computer for information sharing and communication. Computers are most commonly used for:

- Education
- Business
- Engineering
- Medicine
- Scientific research
- Assisting in railway reservation
- Playing games

Main characteristics of a computer:

1. **Speed:** A computer is a very fast device. It can carry out instructions at a very high speed. The speed of a computer is measured in megahertz (MHz) and Giga Hertz (GHz).
2. **Accuracy:** The accuracy of a computer is consistently high. The degree of accuracy of a particular computer depends on the instructions and the type of processor.
3. **Versatility:** Multi-purpose feature of computer makes it quite versatile in nature. One moment, it is doing one job, the next moment it is busy in performing another job.
4. **Storage Capability:** Computers can store huge amounts of data and it can recall the required information immediately.
5. **Diligence:** A computer is free from tiredness and lack of concentration. It can work for hours without creating an error.

History/ Evolution of computers:

In ancient times, people used fingers to perform the calculations such as addition and subtraction. Then sticks, pebbles etc. were used to represent numbers. The development of the modern day computer is the result of advancement in technology. Many inventors have contributed to the history of computers. Few popular early computing devices are discussed below:

Abacus: Abacus was the first calculating device invented around 5000 BC in China. It was used in business as well as in scientific areas such as astronomical calculations, trigonometric calculations, etc. Even in present time, abacus is popular among children to perform calculations.

Napier's Bones: In 1617, John Napier invented a calculating device called Napier's bones. This tool was used for multiplication and division of numbers.

Pascaline: In 1642, Blaise Pascal invented an adding machine called Pascaline. The Pascaline was a decimal machine. This device was used to perform addition and subtraction.

In 1694, Leibnitz improved Pascal's machine so that it could perform multiplication and division too. Such calculators were used throughout the eighteenth century by scientists all over Europe.

Difference Engine: It was designed and partially built during 1821 by Charles Babbage. The difference engine was a huge complex abacus. The Difference Engine could be used to mechanically generate mathematical tables and to solve differential equations. In 1833, Charles Babbage quit working on this machine to concentrate on the analytical engine.

Analytical Engine: In 1834, Charles Babbage designed a machine called the Analytical Engine. It could perform addition, subtraction, multiplication and division automatically. It was the first fully automatic calculating machine. *Charles Babbage* is called the Father of Computers.

Lady Ada Augusta was an assistant to Charles Babbage. She came up with the idea of giving instructions to the computer in a step-by-step manner. *Lady Ada Augusta* is known as the world's first programmer.

Only part of the machine as a trial piece was completed before Babbage's death in 1871. But his analytical engine is viewed as the basic design for modern computer.

Mark-I Computer: The Mark I computer was built at Harvard university in 1944 by Howard Aiken in collaboration with IBM.

The computer was 55 feet long, 8 feet high and weighed 5 tons. It provided vital calculations for the U.S. Navy during World War II. MARK-I was capable of performing addition, subtraction, division and multiplication. However, it was extremely slow, noisy and bulky.

ENIAC: In 1946, John Eckert and John Mauchly of the Moore School of Engineering developed Electronic Numerical Integrator and Calculator (ENIAC) at the University of Pennsylvania. It used electronic vacuum tubes for its internal parts. Its construction started in 1943 and completed in 1946. It occupied about 1800 square feet and used about 18,000 vacuum tubes, weighing almost 50 tons.

UNIVAC: The Universal Automatic Computer (UNIVAC) was the first commercially available electronic computer. It was also the first general-purpose computer, which was designed to handle both numeric and textual information. It was manufactured by the Eckert-Mauchly Corporation in 1951 and its implementation marked the real beginning of the computer era.

Generations of computers:

First Generation:

The period of first generation was 1942-1960. First generation of computer started with using vacuum tubes as the basic components for memory and circuitry for CPU. It used magnetic drums as memory.

Example: IBM 701, MARK II etc.

The main features of First Generation computers are:

- Used Vacuum tube technology
- Input was given through punched cards and paper tapes
- Supported Machine language only
- Were huge sized and required a large room for installation
- Were very costly
- Speed was very slow (milliseconds)
- Generated lot of heat and needed A.C. for cooling
- Were unreliable

Second Generation:

The period of second generation was 1960-1965. This generation used the transistors as main technology. Magnetic cores were used as primary memory. Magnetic tape and magnetic disks were used as secondary memory. The programs were stored in the memory of the computer.

Example: CDC-3600, IBM-7094 etc.

The main features of Second Generation are:

- Use of transistors
- Input was still given through punched cards and paper tapes
- Supported machine and assembly languages
- Smaller in size as compared to First generation computers
- Still very costly
- Faster than first generation computers (microseconds)
- Generated less heat as compared to First generation computers and needed A.C. for cooling
- Reliable as compared to First generation computers

Third Generation:

The period of third generation was 1965-1975. The third generation of computer is marked by the use of Integrated Circuits (IC's) in place of transistors. A single I.C. has many transistors, resistors and capacitors along with the associated circuitry. The IC was invented by *Jack Kilby*. This development made computers smaller in size, reliable and efficient.

Example: CDC-1700, IBM 3600 series, PDP-11/45 etc.

The main features of Third Generation are:

- IC were used
- Supported high level languages
- Keyboards and monitors were used to interact with these computers
- Smaller in size
- Cheaper than second generation computers and were produced commercially.
- Faster (nanoseconds)
- Generated less heat but A.C was still needed
- More reliable

Fourth Generation:

The period of Fourth Generation was 1971-1990. The fourth generation of computers is marked by the use of Large Scale Integrated (LSI) circuits. Then VLSI (Very Large Scale Integration) was used to manufacture microprocessors. Example: Intel-4004, Apple 1 etc.

The main features of Fourth Generation are:

- Microprocessors used
- Supported high level languages
- Very small in size
- Very cheap
- Very high processing speed (picoseconds)
- No A.C. needed
- Portable and reliable
- Large storage capacity

Fifth Generation:

The period of Fifth Generation is 1990-till date. In the fifth generation, the VLSI technology became ULSI (Ultra Large Scale Integration) technology, resulting in the production of microprocessor chips having ten million electronic components. This generation is based on parallel processing hardware and AI (Artificial Intelligence) software. AI is an emerging branch in computer science,

which interprets means and methods of making computers think like human beings

Example: Robotics, Neural networks etc.

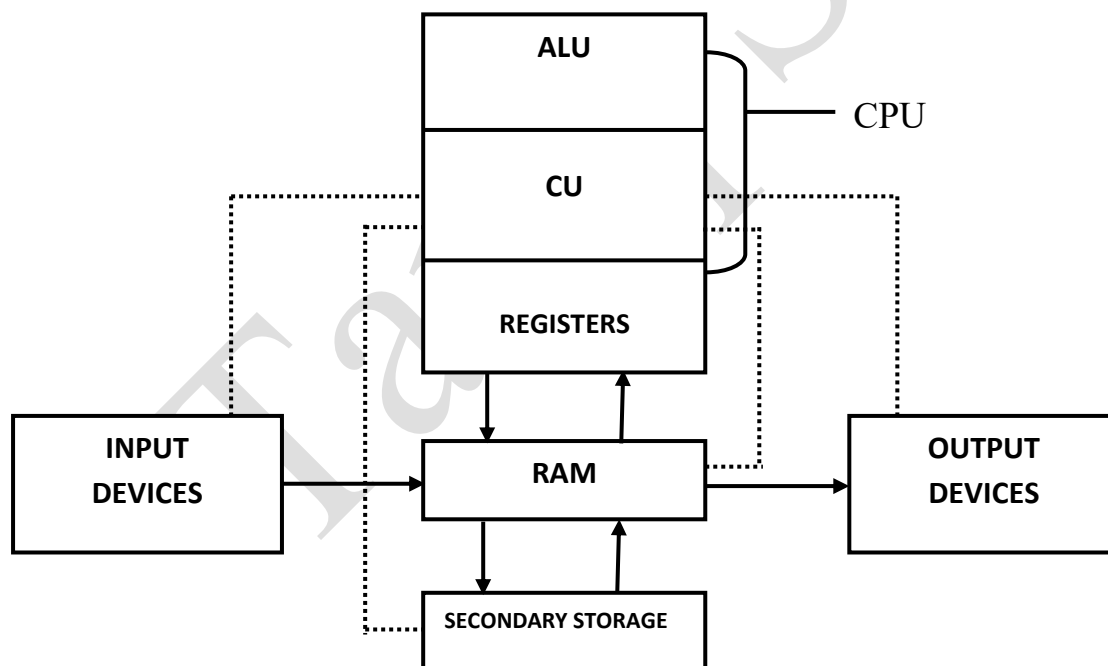
Basic operations/ functions of a computer:

All computer systems perform the following five basic operations:

1. **Input:** The process of entering or providing data and instructions to the computer system.
2. **Storing:** Saving data and instructions so that they are available for processing.
3. **Processing:** Performing arithmetic or logical operations on data as per instructions.
4. **Output:** The process of producing useful information.
5. **Controlling:** Directing the sequence in which all of the above operations are performed.

Q. Draw block diagram of a computer system.

Conventionally a computer consists of input devices, CPU, storage devices and output devices.



Functional components of a computer:

Conventionally a computer consists of input devices, CPU and output devices.

Input Devices: The devices that are used to input data and instructions into the computer are known as input devices. Input devices accept data from the user or some other external source. Data read by the input devices are of different types.

Input devices convert these data and instructions in the binary form and supply the converted form to the computer system.

Examples: keyboard, mouse, scanner, barcode reader, joystick, etc.

Output Devices: The devices used to output the data are called output devices. The result produced by the computer is in binary form. This has to be converted to a human understandable form. This task is carried out by output units.

Example: Monitor printer, plotter, speaker, etc.

Central Processing Unit (CPU): The CPU is known as brain of the computer. All the operations that are performed by a computer are processed by the CPU. It makes all calculations and comparisons. It is responsible for activating and controlling the operations of other units of computer system. It consists of three main components.

- 1. Memory Unit**

It is used to store data and instructions temporarily for use by the processor. It consists of Registers. A register is a small semiconductor memory unit. The registers are of different sizes and capacities viz. 8-bit, 16-bit, and 32-bit, 64 bit, etc.

- 2. Arithmetic-Logic Unit (ALU)**

It is a major component of the central processing unit of a computer system. It performs all the arithmetic and logic operations.

- 3. Control Unit (CU)**

It fetches instructions from programs and directs other units by providing control and timing signals. It issues orders about everything and ensures correct instruction execution. It also transfers data between memory and I/O devices.

Short notes on following devices:

- 1. KEYBOARD:**

Keyboard is the most common and very popular input device. The keyboard is used to input the data to the computer. It is also used to give short cut commands to perform various operations. The layout of the keyboard is like that of traditional typewriter, although there are some additional keys provided for performing some additional functions. The standard arrangement of alphabetic keys is known as the QWERTY keyboard. The characters and symbols are printed on the keys. Keyboards are of two sizes 84 keys or 101/102 keys, but now 104 keys or 108 keys keyboard is also available for Windows and Internet.

- 2. MOUSE:**

The mouse is also an input device. It provides way of interacting with graphical user interfaces on a computer. It can be used to point and click on icons, menus, command buttons etc. It is used to open files and folders. It is also used to run programs and applications. The GUI (Graphical user interface) of current

applications and programs are user friendly and easier to navigate using computer mouse. A mouse generally contains two buttons which have different functions depending on what program is running. A scroll wheel on a mouse is used for scrolling through long documents.

There are two basic mouse types: Mechanical and Optical. These can be defined as follows:

- A. Mechanical mouse:** It has an iron ball coated with rubber, to move in all directions easily. Mechanical sensors within the mouse detect the direction in which the ball is rolling and move the screen pointer accordingly.
- B. Optical mouse:** The optical mouse uses light to track the movements instead of a ball. Optical mouse has a small red light LED that bounces off the light on to a complementary metal oxide semiconductor. The movement is detected by sensing changes in reflected light.

3. Scanner:

A scanner is an input device used to transfer images or text into a computer. There are special models for scanning photo negatives or books. In a computer, the signal from a scanner is transferred to a digital image. This image can then be edited, printed, etc. All scanners have special parts which are used to take a picture from an object. Scanners work in conjunction with computer software programs, which import data from the scanner. Photoshop can edit scanned images. Some programs like Acrobat and OmniPage can actually recognize scanned text.

Types of Scanners:

- A. OMR:** OMR stands for optical mark recognition. OMR technology scans a printed form and reads predefined positions where marks are made on the form. This technology is useful for applications in which large numbers of hand filled forms need to be processed quickly and with great accuracy. A common OMR application is the use of "bubble sheets" for multiple-choice tests used by schools.
- B. OCR:** OCR stands for optical character recognition. It refers to the branch of computer science that involves reading text from paper and translating the images into a form that the computer can manipulate. An OCR system enables you to take a book or a magazine article, feed it directly into an electronic computer file, and then edit the file using a word processor. Most OCR systems use a combination of hardware and software to recognize characters. Advanced OCR systems can read text in large variety of fonts, but they still have difficulty with handwritten text.
- C. MICR:** MICR stands for Magnetic Ink Character Recognition. It is a character recognition system that uses special ink and characters, used primarily by the banking industry to facilitate the processing of cheques and

makes up the routing number and account number at the bottom of a cheque. When a document that contains this ink needs to be read, it passes through a machine, which magnetizes the ink and then translates the magnetic information into characters. It provides a secure, high-speed method of scanning and processing information. The error rate when using MICR cheques is almost zero, which protects consumers and banks from routing funds to the wrong bank or pulling funds from the wrong account.

D. Bar Code Reader (BCR): It is an electronic device for reading and printing line codes and is often used to help organize and index information or prices of objects. These devices are used by the postal service to speed up the delivery of mails. In a super market, barcode scanner reads and logs the prices of a product. It consists of a light source, a lens and a light sensor translating optical impulses into electrical pulses. Lines of different widths and sizes representing a unique code, determines what the scanned object is.

Output devices:

1. VISUAL DISPLAY UNIT (VDU)/ MONITOR:

A VDU (Visual Display Unit) is also called a display screen or monitor. It is an output unit. A monitor is a screen used to display the output. Images are represented on monitors by individual dots called pixels. A pixel is the smallest unit on the screen that can be turned on and off. The density of the dots determines the clarity of the images and is known as the resolution.

Cathode Ray Tubes (CRT): It is a vacuum tube used as a display screen for a computer output device. Although the CRT means only a tube, it usually refers to all monitors.

Liquid-Crystal Displays (LCDs): It is a display technology that creates character by means of reflected light and is commonly used in digital watches and laptop computers. LCDs replaced LEDs (light emitting diodes) because LCDs use less power.

2. PRINTER:

A Printer is an output device. It is used to print documents or images on a paper from a computer. Printed information is called hard copy because the information exists physically. It can also work with digital devices to print directly without the use of a computer.

Printers can be broadly categorized into Impact and Non-impact printers.

A. Impact printers: These printers have a mechanism that involves contact of printer-head (or print hammer or wheel) and the paper to create an image. These printers work by banging a print head containing a number of metal pins which strike an inked ribbon placed between the print head and the paper. Examples are the Dot matrix, Drum, Chain and Line printers.

B. Non -impact Printers: These printers create an image on the print medium without any physical contact between the printer head and the paper. Non-

impact printers are much quieter than impact printers as the head does not strike the paper. An example is an ink-jet and laser printer.

Non-impact printers, used almost everywhere now, are faster and quieter than impact printers because they have fewer moving parts.

Memory and Types of memory:

Memory is the space where data and instructions are stored. The devices used for the storage purpose are called storage devices.

The memory can be classified into two types.

1. Primary Memory
2. Secondary Memory

Primary Memory:

1. It is also known as main memory/ internal memory.
2. It is directly accessible to the CPU.
3. RAM is volatile while as ROM is non-volatile.
4. It has very high speed.
5. It is not suitable for holding large amount of data.
6. It is very costly.

Example: RAM, ROM, PROM etc.

Secondary Memory:

1. It is also known as external memory.
2. It is not directly accessible to the CPU.
3. All secondary storage devices are non-volatile.
4. It is usually slower than primary memory
5. It requires input output channels of the computer.
6. It has higher storage capacity.
7. It is cheaper as compared to primary memory.

Examples: Hard disks, floppy disks, CDs, DVDs, etc.

Cache memory:

Cache memory is a smaller but faster memory used as an intermediate between processor and RAM. The speed of a processor is very high as compared to RAM. Cache memory is intended to provide data to the processor at higher speed. The cache contains a copy of portions of main memory. Cache effectiveness is expressed as a hit ratio. A hit occurs when the data needed by the processor can read it from the cache memory as the copy is already present. A cache miss is when the data needed by the processor is not available in cache memory and hence needs to be accessed from main memory.

Differentiate between RAM and ROM.

RAM:

1. It stands for Random Access Memory.

2. It is volatile.
3. It is used for temporary storage of data and instructions.
4. It can be read as well as written.

ROM:

1. It stands for Read Only Memory.
2. It is non-volatile.
3. It is used for storing programs permanently.
4. It is read only memory and hence cannot be written.

Two types of RAM:

There are primarily two forms of RAM: Static RAM (SRAM) and Dynamic Ram (DRAM).

Static RAM: SRAM retains its contents as long as the power is connected. It is easy to interface. It uses six transistors per bit. It is very expensive. It is faster than its DRAM. Its high price prevents it from being widely used in everyday computing machines, but many modern machines use SRAM as the processor's cache register. SRAM is commonly used in small embedded systems.

Dynamic RAM: DRAM needs regular refresh cycles to prevent its contents from being lost. It is more complicated to interface. It uses only one transistor and a capacitor per bit. It is cheaper per bit. It is slower than SRAM. Widely used in modern computers as primary memory,

Units of measurement of memory:

A computer stores data in units called bits. A bit is the smallest unit of memory capable of storing a 0 or 1.

Other units of memory are

1 Nibble	=	4 Bits
1 Byte	=	8 Bits
1 Kilobyte (1 KB)	= 2^{10} Bytes	= 1024 Bytes
1 Megabyte (1MB)	= 2^{10} KB	= 1024 Kilobytes
1 Gigabyte (1 GB)	= 2^{10} MB	= 1024 Megabytes
1 Terabyte (1 TB)	= 2^{10} GB	= 1024 Gigabytes
1 Petabyte (1PB)	= 2^{10} TB	= 1024 Terabytes
1 Exabyte (1EB)	= 2^{10} PB	= 1024 Petabytes
1 Zettabyte (1ZB)	= 2^{10} EB	= 1024 Exabytes
1 Yottabyte (1YB)	= 2^{10} ZB	= 1024 Zettabytes
1 Brontobyte (1 BB)	= 2^{10} YB	= 1024 Yottabytes
1 Geopbyte (1PB)	= 2^{10} BB	= 1024 Brotobytes

Different variants of ROM:

1. **ROM:** It stands for Read Only Memory. It is non-volatile, hence does not lose information in case of power loss. The storage of data in the ROM is

permanent. It stores some standard programs supplied by the manufacturer. The data on a ROM cannot be changed.

2. **PROM:** It stands for Programmable Read Only Memory. It is also non-volatile. Unlike ROM, it is possible to store programs in a PROM. Once the programs are written, it is not possible to change them again. Which means programs once written cannot be erased.
3. **EPROM:** It stands for Erasable Programmable Read Only Memory. It is also non-volatile. Unlike PROM, it is possible to erase contents of an EPROM chip. So the contents of EPROM chip can be changed by erasing its previous contents. The process of erasing needs the EPROM chip to be exposed to ultraviolet light. Hence it needs to be removed from circuit board for erasing. When the EPROM is in use the information can only be read.
4. **EEPROM:** It stands for Electrically Erasable Programmable Read Only Memory. It is also non-volatile. It can be both erased and programmed with electric pulses. Hence it can be rapidly erased and programmed without removing it from the circuit board.

Short notes on following storage devices:

1. Hard Disk (HDD):

A hard disk drive (HDD) is a secondary storage device used for storing and retrieving digital data. An HDD is non-volatile and hence retains its data even when powered off. It is also called hard drive or hard disk. A hard disk consists of a stack of disk platters that are made up of aluminium alloy of glass coated with a magnetic material. The platters are divided into concentric circles called Tracks. Each Track is further divided into Sectors. Hovering above these platters are read/write heads that transfer data to and from the platters. Data is recorded electromagnetically in sectors on the tracks on the disk. The data on hard drives can be erased and overwritten. Hard drives come with many different storage capacities. The capacity of hard drives is measured in bytes, with common capacities being GB (Giga Bytes) and TB (Tera Bytes). The major manufacturers of the HDD are Seagate, Maxtor, Quantum and Conner Peripherals.

Advantages:

- a. Magnetic disks support direct access of data.
- b. Magnetic disks are suitable for both on-line and off line storage of data.
- c. They can be erased and reused many times.
- d. They are easy to handle and store.
- e. They are easily portable.
- f. Magnetic disk is less vulnerable to data corruption than magnetic tapes.

2. CD:

CD stands for compact disc. It is a digital optical disc storage device. It was co-developed by Philips and Sony. CD comprises of a transparent thin layer of aluminium covering the recording surface for reflectivity. A CD drive uses a low-power laser beam to read digitized (binary) data that has been encoded in the form of tiny pits on an optical disc. The data may be in the form of text, image, sound, video clips, animated movies or their combinations. The diameter of the CD ranges from 10cms to 30cms. A typical CD is of 12cm diameter. One CD can store 600MB to 750MB data on it.

CDs are available in a number of varieties. CD-ROM can only be read but cannot be written or erased. It comes pre-recorded. CD-R can be recorded only once using CD recorder. CD-RW can be recorded and erased a number of times.

3. DVD:

DVD stands for digital versatile disc. It is a type of optical media used for storing digital data. It comes in sizes of 12cm and 8 cm. The 8cm variant is known as Mini-DVD. DVDs have larger storage capacity than CDs. A DVD has a storage capacity of 4.7GB to 17.08GB depending on number of sides and layers used. Some DVDs are formatted specifically for video playback while others may contain different types of data like software applications, music, videos and documents etc.

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4. Pen drives (Flash Drive):

A Pen drive is also called a "Flash drive" or "Thumb drive". It is a portable solid-state data storage device. A Pen drive uses electronic memory called flash memory which is electrically erasable. It is re-writeable and preserves information without a power supply. Flash drives will fit into any USB port on a computer. The drives are quite small and come in a variety of sizes and styles. Pen drives can safely be put into a pocket or purse without fear of damage.

Classification of computers on the basis of size and speed.

Microcomputers:

1. These are also known as personal computers.
2. These computers are usually designed for a single user.
3. They are very fast, but slower than the larger computers.
4. They are smaller in size.
5. They are cheap and user friendly.
6. They constitute the smallest general purpose processing systems.

7. They are used in small businesses, homes, and school classrooms.
8. This category includes desktops, laptops, palmtops etc.

Minicomputers:

1. They are medium sized computers.
2. They are usually designed to serve multiple users.
3. They have larger storage capacity than microcomputers.
4. They are faster than microcomputers.
5. They are expensive as compared to microcomputers.
6. They are used to process large amounts of data for medium sized organizations.
7. They are used as servers in LANs.
8. They are used to control and monitor production processes.

Mainframe Computers:

1. They are big sized computers.
2. They can be used for complex mathematical calculations.
3. They have larger storage capacity than minicomputers.
4. They are faster than minicomputers.
5. They are expensive as compared to minicomputers.
6. They are used by large commercial and government organizations.
7. They are used to process large volumes of data by organizations like banks, insurance companies, railways etc.
8. Qualified and trained operators are required to operate them.

Super computers:

1. They are the largest computers.
2. They are the fastest computers.
3. They are having huge memories and tremendous processing speed.
4. They are very expensive.
5. They are not used for commercial purposes.
6. They are designed for complex scientific applications.

They are used for weather forecasting, space, scientific simulations etc.